EE121 Discussion 5

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Agenda

- Pulse Shaping
- Nyquist Signaling
- Signal Constellations

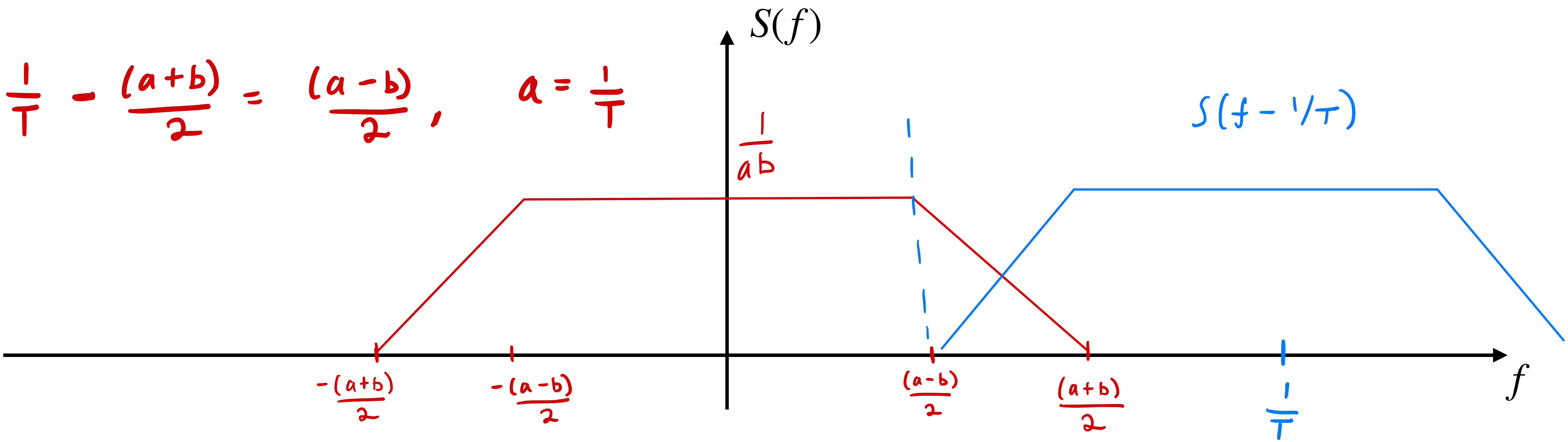
Nyquist Signaling

$S(f) = \frac{1}{a}\Pi(f/a) * \frac{1}{b}\Pi(f/b)$

Consider a pulse $s(t) = \text{sinc}(at)\text{sinc}(bt)$, where $a \geq b$.

(a) Sketch the frequency-domain response $S(f)$ of the pulse.

$\frac{1}{T} - \frac{(a+b)}{2} = \frac{(a-b)}{2}, \quad a = \frac{1}{T}$



Nyquist Signaling

Consider a pulse $s(t) = \text{sinc}(at)\text{sinc}(bt)$, where $a \geq b$.

(b) The pulse is used over an ideal real-baseband channel with one-sided bandwidth 400 Hz. Choose a and b so that the pulse is Nyquist for PAM4 signaling at 1200 bits/s and exactly fills the channel bandwidth.

$\frac{a+b}{2} = 400\text{Hz}$,

4PAM @ 1200 bps is a symbol rate of $\frac{1}{T} = 1200\text{bps} / \log_2 2$ bits/symbol

$\frac{1}{T} = 600$ symbols/sec

$$\boxed{\begin{aligned} a &= 600\text{Hz} \\ b &= 200\text{Hz} \end{aligned}}$$

Nyquist Signaling

Consider a pulse $s(t) = \text{sinc}(at)\text{sinc}(bt)$, where $a \geq b$.

(c) Now, suppose the pulse is used over a passband channel spanning $2.4 - 2.42$ GHz. Assuming that we use 64QAM signaling at 60Mbits/s, choose a and b so that the pulse is Nyquist and exactly fills the channel bandwidth.

Two-sided $\longrightarrow$ $a + b = 20\,\text{MHz}$, $\quad \frac{1}{T} = \frac{60\,\text{Mbps}}{\log_2 64 \text{ bits/symbol}}$

$= 10\,\text{M symbols/sec}$

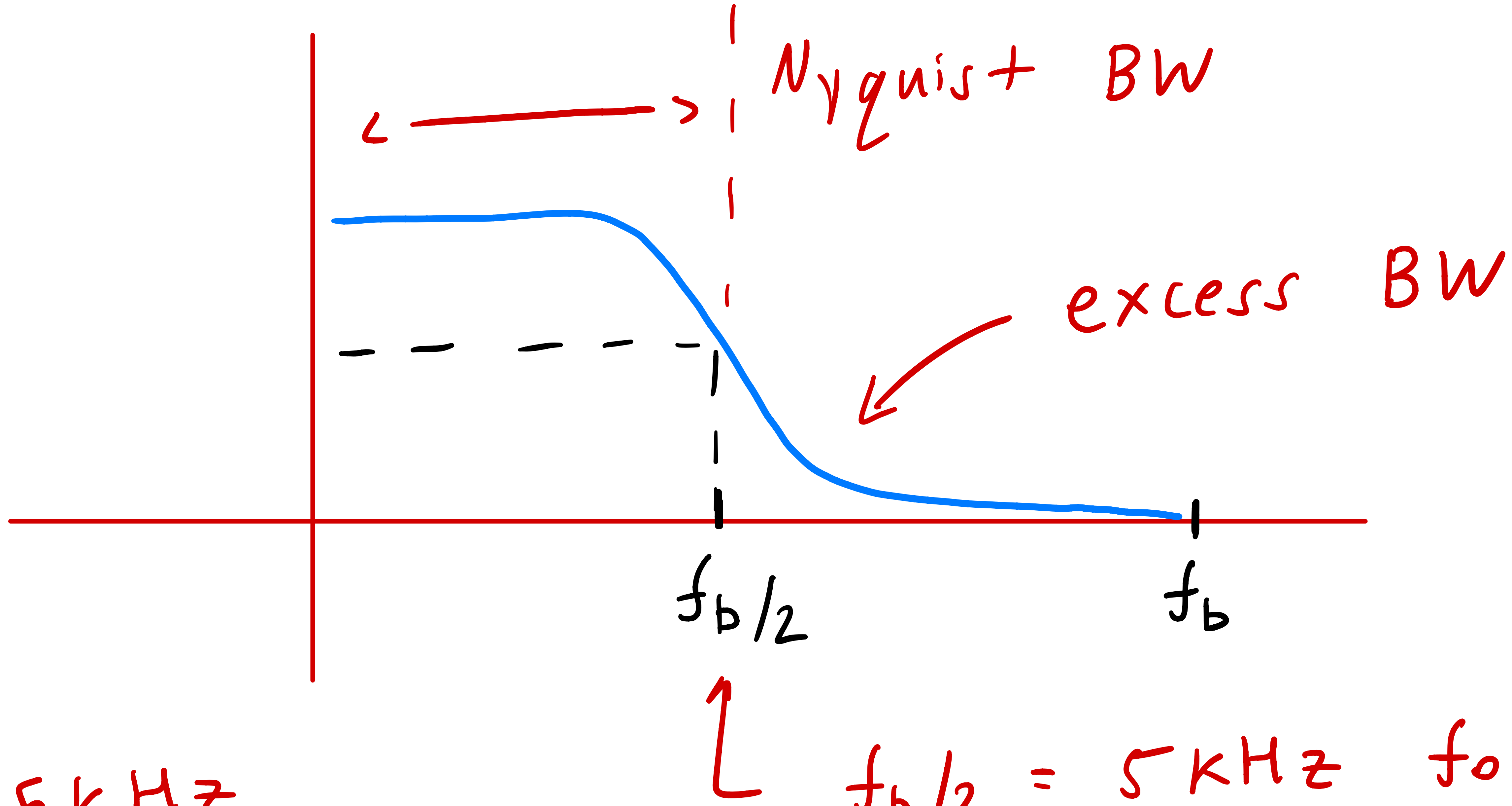
$$\boxed{a = \frac{1}{T} = 10\,\text{MHz}, \quad b = 10\,\text{MHz}}$$

Raised-Cosine Pulse

Assume this is the base band BW

You are given a channel bandwidth of 5 kHz. The requirement is to transmit data over the channel at a rate of 10 kbits/s using binary PAM.

(a) What is the maximum roll-off factor in the raised-cosine pulse spectrum that can accommodate this data transmission?



$$\eta_B = \log_2 M \text{ bits/symbol}$$

$$B_{\min} = \frac{R_b}{\eta_B}$$

$$B = (1+\beta)\frac{R_b}{\eta_B}$$

$f_b/2 = 5\text{ kHz}$ for PAM2
10 kbps

$B = 5\text{ kHz}$
$= (1+\beta) B_{min} = (1+0)\cdot 5\text{ kHz}$

$\boxed{\beta = 0}$

Raised-Cosine Pulse

You are given a channel bandwidth of 5 kHz. The requirement is to transmit data over the channel at a rate of 10 kbits/s using binary PAM.

(b) What is the corresponding excess bandwidth?

$\beta = 0 \longrightarrow BW_{excess} = 0\,Hz$

QPSK Modulator

The binary sequence 11100101 is applied to a QPSK modulator. The bit duration is 1 µs. The carrier frequency is 6 MHz.

(a) Calculate the transmission bandwidth of the QPSK signal.

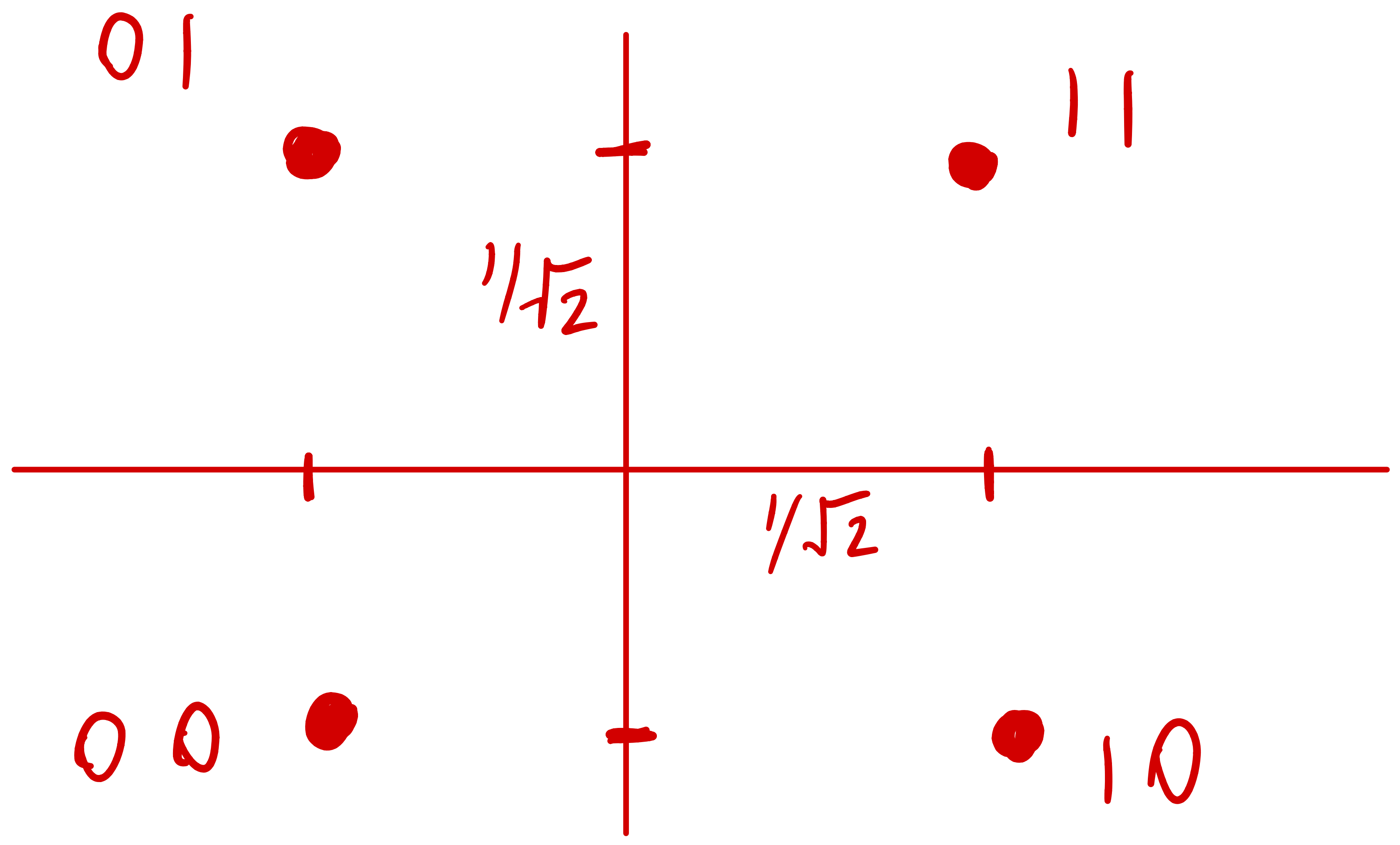
$$B_T = \frac{1}{T_b} = 1\,MHz$$

QPSK Modulator

The binary sequence 11100101 is applied to a QPSK modulator. The bit duration is 1 μs. The carrier frequency is 6 MHz.

(b) Plot the waveform of the QPSK signal.

$$s(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t)$$



	11	10	01	01
$I(t)=$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$
$Q(t)=$	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$